

Tuberculous Lymphadenitis (Scrofula)

Chronic granulomatous inflammation of lymph nodes caused by M. tuberculosis complex, leading to caseation necrosis.

Granuloma

- **Langhans cells:** Multinucleated giant cells found in granulomas.
- **Epithelioid cells:** Transformed macrophages that resemble epithelial cells.

Caseation Necrosis

- **Caseous:** Cheese-like necrotic tissue.

Common Site

- **Sternocleidomastoid muscle:** Neck muscle running diagonally along the side of the neck.

Stages of Tuberculous Lymphadenitis

Stage 1:

- **Lymphoid hyperplasia:** Increase in lymphoid cells causing lymph node enlargement.

Stage 2 & 3:

- **Fluctuating sensation:** Feeling of fluid movement inside nodes.

Stage 4:

- **Fistulae:** Abnormal passages connecting internal areas to the skin surface.

Clinical Features

- **Systemic symptoms:** Signs indicating widespread disease:
 - **Night sweats:** Sweating during sleep due to infections or diseases.

Diagnosis

- **Tuberculin test (PPD):** Skin test to check immune response to TB bacteria.
- **FNAC:** Fine needle aspiration cytology to assess cells in lymph nodes.
- **AFB staining:** Technique to detect acid-fast bacteria like M. tuberculosis.
- **Mycobacterial culture:** Growing TB bacteria in a lab for confirmation.

Treatment

- **HRZE:** Standard first-line TB drugs (Isoniazid, Rifampin, Pyrazinamide, Ethambutol).
- **DOTS:** Directly Observed Treatment, Short-course therapy.

Cystic Hygroma (CH)

Definition

- **Lymphangioma:** Benign tumor of the lymphatic vessels.
 - **Congenital:** Present at birth.
 - **Multiloculated:** Divided into multiple small cavities.
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Common Locations

- **Predilection:** Tendency to occur more commonly on one side (e.g., left side).
 - **Posterior triangle:** Triangular region on the side of the neck behind the sternocleidomastoid muscle.
 - **Axilla:** Underarm region.
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Microscopic Features

- **Locules:** Small cavities or compartments.
 - **Areolar tissue:** Loose connective tissue.
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Clinical Features

- **Transilluminate:** Light passes through the lesion, making it glow.
 - **Doughy masses:** Soft and compressible swellings.
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Associated Complications

- **Airway obstruction:** Blocking of the airway, causing breathing difficulties.
- **Vital structures:** Important anatomical parts like blood vessels and nerves.

Diagnosis

- **MRI/CT Scan:** Imaging techniques to visualize internal structures.
- **Fine Needle Aspiration:** Procedure to extract fluid/cells using a thin needle for analysis.

Treatment

Medical Management:

- **Sclerosing agents:** Substances injected to shrink the lesion.
- **Conservative management:** Non-surgical monitoring and treatment.

Surgical Treatment:

- **Surgical excision:** Complete removal of the lesion.
- **Dissection:** Careful separation of tissues during surgery.

Advanced Options:

- **Laser therapy:** Use of focused light beams to treat the lesion.
- **Radiotherapy:** Treatment using radiation to shrink or control the lesion.

Prognosis

- **Recurrence:** Reappearance of the lesion after removal.

Carotid Body Tumour (CBT)

Definition

- **Chemodectoma/Paraganglioma:** Tumour arising from chemoreceptor cells or paraganglia.
- **Adventitia:** Outer layer of a blood vessel.
- **Carotid bifurcation:** Point where the common carotid artery splits into internal and external branches.

Types of CBT

- **Familial:** Runs in families due to genetic factors.
- **Sporadic:** Occurs randomly, without a family history (most common).
- **Hyperplastic:** Increased size due to excessive cell growth.

Aetiology

- **Hypoxic stimulation:** Prolonged low oxygen levels.
- **Genetic predisposition:** Inherited tendency to develop certain diseases.

Clinical Features

- **Anterior triangle:** Area on the side of the neck bounded by muscles and the lower jaw.
- **Fontaine sign:** Tumour fixed vertically because of its connection to the carotid artery.
- **Bruit:** Whooshing sound caused by turbulent blood flow.

Cranial nerve palsy: Damage to specific cranial nerves causing symptoms:

- **Hypoglossal nerve:** Affects tongue movement.
- **Glossopharyngeal nerve:** Impairs swallowing and taste.
- **Recurrent laryngeal nerve:** Causes hoarseness.
- **Spinal accessory nerve:** Leads to shoulder drooping.
- **Sympathetic chain:** Affected in Horner syndrome, causing:
 - **Ptosis:** Drooping eyelid.
 - **Miosis:** Constricted pupil.
 - **Anhidrosis:** Lack of sweating.

Symptoms Associated with CBTs

- **Dysphagia:** Difficulty in swallowing.
 - **Horner syndrome:** Combination of ptosis, miosis, and anhidrosis.
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Treatment

- **Surgery:** Removal of the tumour.
- **Radiotherapy:** Use of radiation to shrink or control the tumour.

Recurrent Laryngeal Nerve (RLN) Injury

Definition

- **Recurrent Laryngeal Nerve (RLN):** Branch of the vagus nerve controlling vocal cords and laryngeal sensations.
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Causes

- **Thyroid surgery:** Surgery involving the thyroid gland.
 - **Carotid endarterectomy:** Procedure to remove plaque from the carotid artery.
 - **Penetrating wounds:** Deep injuries caused by sharp objects.
 - **Intubation:** Insertion of a tube into the airway for ventilation.
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Symptoms

- **Hoarseness:** Raspy or weak voice.
 - **Stridor:** High-pitched noise during breathing caused by airway obstruction.
 - **Aspiration:** Food or liquid entering the airway instead of the esophagus.
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Diagnosis

- **Laryngoscopy:** Procedure to visualize the vocal cords.
 - **Electromyography (EMG):** Test to measure electrical activity of muscles and nerves.
 - **CT/MRI:** Imaging techniques to detect structural abnormalities or tumors.
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Management

- **Thyroplasty:** Surgical procedure to improve vocal cord function.
 - **Vocal cord injection:** Injection of substances to reposition or strengthen vocal cords.
 - **Reinnervation surgery:** Restores nerve supply to vocal cords.
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Prevention

- **Intraoperative nerve monitoring:** Use of technology during surgery to avoid nerve damage.
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Complications

- **Aspiration pneumonia:** Lung infection caused by inhaling food or liquid.
- **Tracheostomy:** Surgical procedure to create an opening in the airway for breathing.

Tracheostomy

Definition

- **Surgical airway:** Artificial opening into the trachea to assist breathing.
 - **Cervical trachea:** Part of the trachea located in the neck.
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Indications

- **Bypass obstruction:** Creating an alternate airway path.
 - **Congenital anomalies:** Birth defects like underdeveloped larynx (laryngeal hypoplasia).
 - **Vascular web:** Abnormal blood vessel network causing obstruction.
 - **Supraglottic/glottic conditions:** Issues above or at the vocal cords.
- **Subcutaneous emphysema:** Air trapped under the skin.
- **Facial fractures:** Broken bones in the face blocking the airway.
- **Pulmonary toilet:** Clearing lung secretions effectively.
- **CPAP:** Continuous Positive Airway Pressure, a non-invasive ventilation method.

Tracheostomy Procedure

Position

- **Supine:** Lying on the back with the face upward.
- **Semi-recumbent:** Halfway between lying down and sitting up.

Landmarks

- **Thyroid notch:** Small depression on the upper border of the thyroid cartilage.
- **Sternal notch:** Depression at the top of the sternum.
- **Cricoid cartilage:** Ring-shaped cartilage below the thyroid cartilage.

Incision

- **Vertical incision:** Straight cut along the midline.
- **Horizontal incision:** Side-to-side cut for better appearance.

Dissection

- **Platysma:** Thin muscle under the skin of the neck.
- **Strap muscles:** Small muscles in the front of the neck.
- **Pretracheal fascia:** Connective tissue in front of the trachea.
- **Thyroid isthmus:** Bridge of tissue connecting the two lobes of the thyroid gland.
- **Innominate artery:** Major artery in the chest, near the trachea.

Tracheal Opening

- **T-shaped incision:** Horizontal cut between tracheal rings for access.
- **U- or H-shaped incision:** Reflecting flaps for better access, especially in obese patients.
- **Permanent stoma:** Long-term opening in the trachea for breathing.

Procedure Completion

- **Suction:** Removing blood or secretions from the airway.
 - **Tracheostomy tube:** Tube inserted to maintain the airway.
 - **Carbon dioxide return:** Confirms proper placement by detecting exhaled CO₂.
 - **Tracheostomy collar:** Device used to secure the tube.
 - **Subcutaneous emphysema:** Air leakage under the skin, prevented by leaving the skin open.
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Post-procedure Management

- **Iodine gauze:** Sterile dressing to prevent infection.
- **Petrolatum gauze:** Oily dressing to manage oozing and keep the wound moist.

Postoperative Management and Complications of Tracheostomy

Postoperative Management

- **Irrigation:** Flushing the trachea with saline to clear secretions.
 - **Suctioning:** Removal of secretions or mucus from the airway.
 - **Humidified oxygen:** Oxygen with added moisture to prevent dryness in the airway.
 - **Mucolytic agents:** Medications that break down mucus, making it easier to clear.
 - **Cannula:** Tube inserted into the trachea to maintain the airway.
 - **Downsize:** Reducing the size of the tube as the patient improves.
 - **Sutures:** Stitches used to close the wound after surgery.
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Complications of Tracheostomy

A) Immediate Complications

- **Apnoea:** Temporary cessation of breathing, often due to loss of respiratory drive.
- **Vascular thyroid gland:** Blood vessels associated with the thyroid that can bleed.
- **Pneumothorax:** Air in the chest cavity, causing lung collapse.
- **Pneumomediastinum:** Air in the space between the lungs.
- **Recurrent laryngeal nerves:** Nerves controlling the vocal cords, which can be injured during surgery.
- **Postobstructive pulmonary oedema:** Fluid accumulation in the lungs after the airway is cleared.
- **Endotracheal tube ignition:** Fire in the tube, typically from the use of electrocautery or laser.

B) Early Complications

- **Plugging with mucus:** Blockage of the tube by mucus.
- **Tracheitis:** Inflammation or infection of the trachea.
- **Cellulitis:** Skin infection around the surgical site.
- **Displacement of the tube:** The tube moving out of place.
- **Subcutaneous emphysema:** Air trapped under the skin due to tube placement issues.

- **Atelectasis:** Collapse of lung tissue, often caused by improper tube placement.

C) Late Complications

- **Tracheomalacia:** Weakening and softening of the tracheal walls.
- **Tracheo-oesophageal fistula:** Abnormal connection between the trachea and the oesophagus.
- **Tracheocutaneous fistula:** Abnormal connection between the trachea and the skin.
- **Granulation tissue:** Overgrowth of tissue at the site of the incision or injury.

Quinsy (Peritonsillar Abscess)

Clinical Features

History

- **Malaise:** A general feeling of discomfort or illness.
- **Dysphagia:** Difficulty swallowing.
- **Odynophagia:** Painful swallowing.
- **Otalgia:** Ear pain.

Physical Evaluation

- **Trismus:** Painful or limited jaw movement.
 - **Hot potato/muffled voice:** A sound like talking with a potato in the mouth, due to swelling and abscess.
 - **Rancid or fetid breath:** Bad or foul-smelling breath, often due to infection.
 - **Cervical lymphadenitis:** Swelling of the lymph nodes in the neck.
 - **Asymmetric tonsillar hypertrophy:** Enlargement of the tonsil on one side.
 - **Localized fluctuance:** Swelling with a fluid-filled space, indicative of abscess formation.
 - **Inferior and medial displacement:** The tonsil moving downwards and towards the middle.
 - **Contralateral deviation of the uvula:** Uvula shifting to the opposite side of the infected tonsil.
 - **Erythema:** Redness, usually due to inflammation.
 - **Exudates:** Pus or fluid that drains from an infected area.
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Treatment

- **Needle aspiration:** Inserting a needle to remove fluid from the abscess.
- **Incision and drainage:** Surgical procedure to drain the abscess.
- **Empiric antibiotics:** Antibiotics prescribed based on the most likely pathogens, before specific identification.
- **Clindamycin:** An antibiotic used for bacterial infections.
- **Antipyretics:** Medications used to reduce fever.
- **Analgesics:** Pain-relieving medications.
- **Endotracheal intubation:** Inserting a tube into the airway to secure breathing.
- **Cricothyroidotomy:** Surgical procedure to create an airway opening in the neck.
- **Tracheotomy:** Surgical procedure to create an opening in the trachea for long-term airway management.

Collar Stud Abscess

Pathophysiology

- **Suppurative infection:** Infection characterized by the formation of pus.
- **Blister:** A small bubble of fluid that forms on the skin due to irritation or infection.
- **Neurovascular bundle:** A group of nerves and blood vessels that travel together in the body.
- **Deep fascia:** A connective tissue layer beneath the skin that surrounds muscles and other structures.
- **Subcutaneous plane:** The layer of tissue located beneath the skin, containing fat and connective tissue.
- **Cold abscess:** A pus-filled swelling that forms slowly and without much heat or redness.

Management

- **Incision:** A surgical cut made to allow drainage.
 - **Delayed closure:** The technique of leaving the wound open for some time before closing it, to allow for drainage and prevent pus from being trapped inside.
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Treatment

- **Systemic antibiotics:** Antibiotics taken orally or intravenously to treat infections throughout the body.
- **Pathogen:** The microorganism (bacteria, virus, etc.) causing the infection.
- **Necrosis:** Death of tissue, often due to infection or lack of blood supply.
- **Sepsis:** A life-threatening response to infection that can spread throughout the body.

Anatomy of Thyroid Gland

Location

- **Thyroid cartilage:** The cartilage that forms the front of the larynx (voice box).
 - **Trachea:** The windpipe, which carries air to the lungs.
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Shape

- **Isthmus:** A narrow strip of tissue that connects two larger structures.
 - **Pyramidal lobe:** An extra lobe of the thyroid, a remnant of embryonic tissue.
 - **Thyroglossal duct:** An embryonic structure that usually disappears after birth but may leave behind remnants.
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Coverings

- **Fibrous capsule:** A tough layer of connective tissue that surrounds the thyroid gland.
 - **Pretracheal fascia:** A layer of connective tissue located in front of the trachea.
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Relations

- **Strap muscles:** Muscles that lie in the front of the neck, involved in movement and support.
- **Recurrent laryngeal nerve:** A nerve that controls the muscles of the voice box.
- **Common carotid artery:** A major artery in the neck that supplies blood to the head and neck.

- **Internal jugular vein:** A vein that drains blood from the brain, face, and neck.
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Blood Supply

- **Arterial:** Relating to arteries, which carry blood from the heart.
 - **Superior thyroid artery:** Supplies the upper part of the thyroid gland.
 - **Inferior thyroid artery:** Supplies the lower part of the thyroid gland.
 - **Venous:** Relating to veins, which carry blood back to the heart.
 - **Superior and middle thyroid veins:** Drain the upper and middle parts of the thyroid gland.
 - **Inferior thyroid vein:** Drains the lower part of the thyroid gland.
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Nerve Supply

- **Sympathetic:** Part of the nervous system that controls the body's fight or flight response.
 - **Parasympathetic:** Part of the nervous system that controls rest and digestion.
 - **Vagus nerve:** A major nerve that controls functions like heart rate and digestion.
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Lymphatic Drainage

- **Lymph nodes:** Small, bean-shaped structures that filter lymph and store white blood cells.
 - **Deep cervical and paratracheal lymph nodes:** Lymph nodes in the neck that drain the thyroid gland.
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Functions

- **Thyroid hormones (T3 and T4):** Hormones that regulate metabolism (how the body uses energy).
 - **Calcitonin:** A hormone that helps regulate calcium levels in the blood.
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Clinical Importance

- **Goiter:** Enlargement of the thyroid gland.
- **Thyroidectomy:** Surgical removal of the thyroid gland.
- **Thyroid nodules:** Abnormal growths in the thyroid, which can be benign (non-cancerous) or malignant (cancerous).

Peripheral Nerve Injury

Definition

- **Peripheral nerves:** Nerves outside the brain and spinal cord that transmit signals to and from the limbs and organs.
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Causes

- **Trauma:** Physical injury like cuts, crushes, or stretching of nerves.
 - **Compression:** When a nerve is pinched or squeezed by a tumor, hematoma (blood clot), or tight cast.
 - **Iatrogenic:** Injuries caused by medical treatments or procedures, such as surgery.
 - **Infections:** Conditions like leprosy that affect the nerves.
 - **Metabolic disorders:** Conditions like diabetes or alcoholism that can cause nerve damage.
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Types of Nerve Injuries

- **Neurapraxia:** A temporary block of nerve conduction without structural damage.
 - **Axonotmesis:** Damage to the nerve fibers (axons) while the nerve sheath remains intact.
 - **Neurotmesis:** Complete disruption of both the nerve fibers and the nerve sheath.
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Clinical Features

- **Motor Symptoms:** Weakness or paralysis (loss of muscle movement).
- **Sensory Symptoms:** Numbness, tingling, or loss of sensation.

- **Autonomic Symptoms:** Loss of functions like sweating or temperature regulation, which are controlled automatically by the nervous system.
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Diagnosis

- **Clinical Examination:** Testing muscle strength, reflexes, and sensation to assess nerve function.
 - **Electromyography (EMG):** A test that measures the electrical activity of muscles and nerves.
 - **Nerve Conduction Studies (NCS):** Tests to measure the speed at which electrical signals travel along the nerves.
 - **Imaging:** MRI or ultrasound used to identify nerve compression or damage.
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Management

- **Conservative:** Non-surgical treatments like rest, physiotherapy, and pain management.
 - **Surgical:** Procedures to repair, graft, or decompress nerves.
 - **Rehabilitation:** Muscle re-education and occupational therapy to restore function.
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Complications

- **Chronic pain:** Long-lasting pain, often described as burning or shooting, caused by nerve damage (neuropathic pain).
 - **Permanent paralysis:** Loss of muscle function due to nerve damage.
 - **Contractures:** Muscle shortening or joint stiffness due to imbalance in muscle strength after nerve injury.
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Prognosis

- The outcome depends on the **type** of injury, the **extent** of nerve damage, and the **timing** of medical intervention.

Mixed Parotid Tumor (Pleomorphic Adenoma)

Definition

- **Pleomorphic Adenoma:** A benign tumor of the parotid gland that consists of both epithelial and mesenchymal (connective tissue) components.
 - **Aneuploid:** A condition where the number of chromosomes is abnormal.
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Components

- **Epithelial cell component:** The tissue formed by the cells that line the ducts in the gland.
 - **Myoepithelial cell component:** The tissue made up of contractile cells that are involved in the secretion process.
 - **Stromal (mesenchymal) component:** The supporting tissue in the tumor, often composed of connective tissue.
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Common Sites

- **Tail of superficial lobe of parotid gland:** The most common site, affecting 70-80% of cases.
 - **Submandibular gland:** Occurs in 10% of cases.
 - **Sublingual gland:** Rarely affected.
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Clinical Features

- **Solitary:** Usually a single mass.
- **Slow-growing:** The tumor grows gradually over time.
- **Firm, mobile, painless, nodular mass:** The tumor is generally firm to touch, movable, and doesn't cause pain.
- **Colour varies:** Can be whitish, tan, grey, or bluish.
- **Size:** Ranges from a few millimeters to very large.
- **Mobile, unless located in the palate:** Typically movable unless located in certain areas like the palate.
- **Ear lobe eversion:** When the tumor is located in the parotid tail, it can cause the ear lobe to protrude.
- **Facial nerve weakness:** Rarely occurs unless the tumor is large or malignant.
- **Oral retrotonsillar mass:** In the deep lobe of the parotid, the tumor can appear as a mass in the back of the throat or parapharyngeal space.

- **Minor salivary gland tumors:** Can cause symptoms like difficulty swallowing (dysphagia), difficulty breathing (dyspnoea), hoarseness, chewing difficulty, and nosebleeds (epistaxis).
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Diagnosis

- **Tissue Sampling Procedures**
 - **FNA (Fine Needle Aspiration):** A needle is used to take a small sample of tissue; sensitivity is around 90%.
 - **Core Needle Biopsy:** A larger needle is used to obtain a bigger tissue sample; sensitivity is around 97%.
 - Recommended to perform before surgery to determine tumor characteristics.
 - **Diagnostic Imaging Techniques**
 - **Ultrasound:** Helpful for visualizing superficial tumors.
 - **CT Scan:** Useful for determining tumor size and checking for bone involvement.
 - **MRI:** More effective for identifying soft tissue involvement and perineural (nerve) invasion.
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Treatment

- **Surgical Resection:** The main treatment for pleomorphic adenoma.
 - **For Parotid Gland**
 - **Superficial Parotidectomy:** Removal of part of the parotid gland.
 - **Total Parotidectomy:** Preferred to reduce the chance of recurrence.
 - **Preserve Facial Nerve:** Effort is made to avoid damage to the facial nerve during surgery.
 - **For Submandibular Gland**
 - **Simple Excision:** The tumor is removed with care to preserve important nerves like the mandibular branch of the trigeminal nerve, hypoglossal nerve, and lingual nerve.
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Acute Parotitis: Causes, Clinical Features, and Treatment

Aetiology of Acute Parotitis

1. **Acute Bacterial Parotitis:** Bacterial infection causing inflammation of the parotid glands.
 2. **Acute Viral Parotitis (Mumps):** Inflammation of the parotid glands caused by the mumps virus.
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Acute Bacterial Parotitis

- **Staphylococcus aureus:** A type of bacteria commonly responsible for causing infections.
 - **Commensal bacteria:** Bacteria that normally live in the body without causing harm, but can cause infection in certain conditions.
 - **Postoperative or severely ill patients:** Acute bacterial parotitis is more common in patients recovering from surgery or those who are critically ill.
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Acute Viral Parotitis (Mumps)

- **Paramyxovirus:** The virus that causes mumps.
 - **MMR vaccine:** A vaccine that protects against measles, mumps, and rubella.
 - **1st dose at 1 year:** The first dose of the vaccine is given at 1 year of age.
 - **2nd dose at 4-6 years:** The second dose is given between 4-6 years of age.
 - **Bilateral involvement:** A condition where both parotid glands are affected.
 - **Malaise:** A general feeling of discomfort or illness.
 - **Orchitis:** Inflammation of the testes, a possible complication of mumps.
 - **Deafness:** Loss of hearing, which can occur as a complication in young adults.
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Clinical Features

- **Acute bacterial parotitis:**
 - **Progressive painful swelling:** The swelling of the gland increases over time and becomes painful.
 - **Purulent saliva:** Thick, yellowish or greenish saliva that is pus-like.
- **Acute viral parotitis (mumps):**
 - **Pain and swelling:** Pain and swelling last for 5-9 days, often affecting both glands.
 - **Clear saliva:** Saliva that is normal in color when expressed from the glands.

Treatment

- **Acute bacterial parotitis:**
 - **Empirical therapy:** Initial treatment before test results are available.
 - **Vancomycin:** An antibiotic used to treat infections caused by resistant bacteria.
 - **Methicillin-resistant Staphylococcus aureus (MRSA):** A type of Staphylococcus aureus resistant to some antibiotics, requiring special treatment.
 - **Investigations:** Tests like Gram stain and culture to identify the bacteria causing the infection.
 - **Culture results:** Test results that help tailor treatment based on the specific bacteria involved.
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Submandibular Salivary Calculus (Sialolithiasis): Pathology, Clinical Features, and Management

Reasons for Higher Frequency of Calculi Formation in Submandibular Gland

- **pH of saliva:** The saliva in the submandibular gland is more alkaline (basic), which can contribute to calculus formation.
 - **Viscosity of saliva:** The saliva in the submandibular gland is thicker (more mucous), which can increase the likelihood of stasis and stone formation.
 - **Anatomical factors:** Wharton's duct has an uphill course in the submandibular gland, making it more prone to obstruction and stone formation.
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Common Location

- **Wharton's duct:** The duct that drains saliva from the submandibular gland, often the site where stones form, especially near its orifice or at the bend behind the mylohyoid muscle.
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Pathology

- **Increased viscosity:** Due to the higher concentration of mucus in the submandibular saliva, it becomes thicker and flows more slowly, which can lead to stagnation and the formation of calculi.
 - **Stasis:** The reduced flow of saliva leads to stasis (sluggish or no movement), which can cause stone formation.
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Clinical Features

- **Palpable lump or visible swelling:** The stone in the submandibular gland may cause a lump or swelling in the area.
 - **Pain:** Pain usually originates from the floor of the mouth, where the submandibular gland is located.
 - **Intermittent swelling:** Swelling can occur during meals due to increased salivation.
 - **Complications:** If obstruction persists, it can lead to bacterial infections and sialadenitis (inflammation of the salivary gland).
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Treatment

1. **Small stones:**
 - **Hydration:** Drinking plenty of water helps keep the gland flushed.
 - **Moist heat:** Applying heat can help promote salivation and stone expulsion.
 - **NSAIDs:** Nonsteroidal anti-inflammatory drugs to relieve pain and inflammation.
 - **Sucking on sour foods:** Citrus fruits can stimulate salivation and may help in expelling small stones.
 - **Massage:** In some cases, gentle massage by a specialist can help remove the stone.
 2. **For larger stones:**
 - **Duct cannulation and stone removal:** A procedure where a small tube is inserted into the duct to remove the stone.
 - **Gland excision:** Removal of the submandibular gland in cases of recurrent or severe stones.
 - **Sialendoscopy:** A minimally invasive technique to remove the stone using a small camera and instruments inserted into the duct.
 3. **Antibiotics:** For treating bacterial infection if sialadenitis occurs.
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Frey's Syndrome: Meaning of Hard Words

Symptoms

- **Redness:** Increase in blood flow, causing the skin to appear more flushed or pink.
 - **Sweating:** The process of producing sweat, often as a response to eating or thinking about food.
 - **Gustatory sweating:** Sweating specifically triggered by the act of eating, particularly foods that stimulate saliva production.
 - **Lemon wedges:** A type of citrus fruit that, when eaten, can trigger the characteristic sweating associated with Frey's syndrome.
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Cause

- **Auriculotemporal branch:** A branch of the trigeminal nerve responsible for transmitting sensory signals from the face and also carrying parasympathetic and sympathetic nerve fibers.
 - **Trigeminal nerve:** A large nerve responsible for sensation in the face and motor functions such as chewing.
 - **Parasympathetic fibres:** Nerve fibers that usually promote saliva production by the parotid gland.
 - **Sympathetic fibres:** Nerve fibers that primarily control the body's fight-or-flight response, including stimulating sweat production.
 - **Regeneration:** The process of nerve fibers growing back after injury. In Frey's syndrome, nerve fibers regenerate inappropriately, causing unwanted effects.
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Treatment Options

- **Botulinum toxin type A:** A neurotoxin used to block nerve signals temporarily, which can help reduce sweating.
- **Surgical transection:** Cutting or severing the nerve fibers to prevent the abnormal stimulation of sweating.
- **Anticholinergic ointment:** Creams or ointments containing substances like scopolamine that can block the action of acetylcholine, a neurotransmitter involved in sweat production.